3 **(a)** The equation $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ can be used in calculations to determine the internal energy of a gas.

(i) State the meaning of the following:

1. m .....

2. $\langle c^2\rangle$ .....

.....

3. $\frac{1}{2}m\langle c^2\rangle$ .....

.....

[3]

(ii) Calculate the internal energy of one mole of an ideal gas at 27.0 °C.

internal energy = ... J [2]

(b) The atmosphere is mostly a mixture of nitrogen and oxygen molecules.

For molecules in the atmosphere, deduce the ratio

$$\frac{\text{r.m.s. speed of a nitrogen molecule}}{\text{r.m.s. speed of an oxygen molecule}}.$$

Mass of a nitrogen molecule = 4.65×10^{-26} kg.
Mass of an oxygen molecule = 5.31×10^{-26} kg.

ratio = ... [2]

(c) During the first part of the power stroke in a diesel engine, the pressure of the burning gas remains constant at 8.00×10^6 Pa, while the volume of the gas increases from 3.00×10^{-5} m^3 to 7.10×10^{-5} m^3.

(i) Calculate the work done by the gas during the first part of the power stroke.

work done = ... J [2]

(ii) While this expansion is taking place, the temperature of the gas rises by 900 K. The mass of the gas is 1.27×10^{-3} kg and its specific heat capacity at constant pressure is 1.01×10^3 J kg^{-1} K^{-1}.

Calculate the heat gained by the burning gas.

heat gained = ... J [2]

(iii) Deduce, using your answers to **3(c)(i)** and **3(c)(ii)**, the change in the internal energy of the burning gas.

change in internal energy = ... J [1]

[Total: 12]